

HDOC DAY-7

Q1. Write a C program to find echo distance when speed of sound and time are given.

First we will form a Test case table

Test case	Inputs	Expected output	Actual output	Result
1	340, 3	510 m	510 m	pass
2	330, 5	825 m	825 m	pass
3	350, 7	1225 m	1225 m	pass

Now that we have made the test case table we will write the algorithm that we are going to use

1. Start
2. Declare variables speed, time, distance.
3. Read the values of speed and time
4. Calculate distance = (speed*time)/2
5. Display distance
6. Stop

Now we will write the source code

```
#include <stdio.h>
```

```
Int main()
```

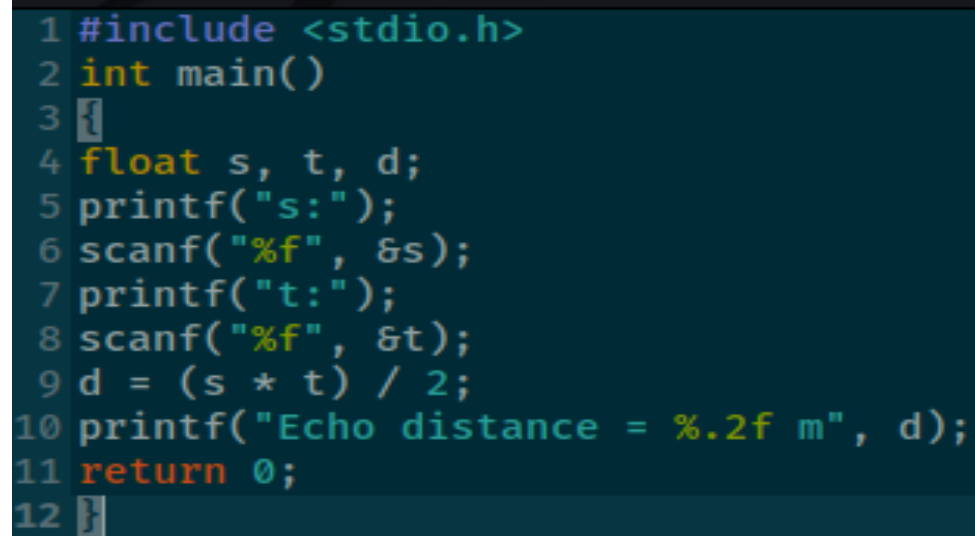
```
{
```

```
float s, t, d;
```

```
printf("s:");
```

```
scanf("%f", &s);  
printf("t:");  
scanf("%f", &t);  
d = (s * t) / 2;  
printf("Echo distance = %.2f m", d);  
return 0;  
}
```

Now we will run the test cases in our source code:



```
1 #include <stdio.h>  
2 int main()  
3 {  
4     float s, t, d;  
5     printf("s:");  
6     scanf("%f", &s);  
7     printf("t:");  
8     scanf("%f", &t);  
9     d = (s * t) / 2;  
10    printf("Echo distance = %.2f m", d);  
11    return 0;  
12 }
```

```
Session Actions Edit View Help
(kali㉿kali)-[~]
$ gedit hdoc7.1.c

** (gedit:3083): WARNING **: 03:58:50.288: Could not
** (gedit:3083): WARNING **: 03:58:50.288: Could not

(kali㉿kali)-[~]
$ cc -g hdoc7.1.c

(kali㉿kali)-[~]
$ ./a.out
s:340
t:3
Echo distance = 510.00 m

(kali㉿kali)-[~]
$ ./a.out
s:330
t:5
Echo distance = 825.00 m

(kali㉿kali)-[~]
$ ./a.out
s:350
t:7
Echo distance = 1225.00 m
```

Q.2. Write a C program to find heat transfer rate when thermal conductivity, area, temperature difference, and thickness are given.

We will do the same steps as we did in the first question

Test case table

Test Case	Input(k,A,T,L)	Expected output	Actual output	Result
1	200, 2, 40, 0.5	32000 W	32000 W	pass
2	150, 1.5, 30, 0.3	22500 W	22500 W	pass
3	100, 3, 20, 0.4	15000 W	15000 W	pass

Now we will make the Algorithm

1. Start
2. Declare variables K, A, T, L, heatrate.
3. Read values of K, A, T, L
4. Calculate $\text{heatrate} = (k \times A \times T) / L$
5. Display heatrate
6. Stop

Now we will write the code:

```
#include <stdio.h>

Int main()
{
float k, A, T, L, heatrate;
printf("K:");
scanf("%f", &K);
printf("A:");
scanf("%f", &A);
printf("T:");
scanf("%f", &T);
printf("L:");
scanf("%f", &L);
heatrate = (K x A x T) / L;
printf("heat rate transfer = %.2f W", heatrate);
return 0;
```

}

Now we will execute and compile the code:

```
1 #include <stdio.h>
2
3 int main()
4 {
5     float K, A, T, L, heatrate;
6
7     printf("K:");
8     scanf("%f", &K);
9
10    printf("A:");
11    scanf("%f", &A);
12
13    printf("T:");
14    scanf("%f", &T);
15
16    printf("L:");
17    scanf("%f", &L);
18
19    heatrate = (K * A * T) / L;
20
21    printf("Heat Transfer Rate = %.2f W", heatrate);
22
23    return 0;
24 }
```

```

(kali㉿kali)-[~]
$ gedit hdoc7.2.c

** (gedit:28996): WARNING **: 04:43:22.598:
** (gedit:28996): WARNING **: 04:43:22.598:
gedit: hdoc7.2.c
(kali㉿kali)-[~]
$ cc -g hdoc7.2.c

(kali㉿kali)-[~]
$ ./a.out hdoc7.2.c
K:200
A:2 (gedit:22502): WARNING **: 04:43:55.922:
T:40
L:0.5 (gedit:22502): WARNING **: 04:43:55.922:
Heat Transfer Rate = 32000.00 W
(kali㉿kali)-[~]
$ ./a.out
K:150 (kali㉿kali)-[~]
A:1.5
T:30
L:0.3 (kali㉿kali)-[~]
Heat Transfer Rate = 22500.00 W

(kali㉿kali)-[~]
$ ./a.out
K:100
A:3
T:20
L:0.4
Heat Transfer Rate = 15000.00 W

```

Q.3 Write a C program to find transformer secondary voltage when primary voltage, primary turns, and secondary turns are given.

Test case	Input(Vp, Np, Ns)	Expected output	Actual output	Result
1	220, 500, 250	110V	110V	Pass
2	230, 1000, 500	115V	115V	Pass
3	240, 800, 400	120V	120V	Pass

Algorithm:

- 1 Start.
- 2 Declare primaryVoltage, primaryTurns, secondaryTurns and secondaryVoltage as float.
- 3 Read primaryVoltage, primaryTurns and secondaryTurns.
- 4 Calculate $\text{secondaryVoltage} = (\text{primaryVoltage} \times \text{secondaryTurns}) / \text{primaryTurns}$.
- 5 Display secondaryVoltage.
- 6 Stop.

Now we will write the code

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    float primaryVoltage, primaryTurns, secondaryTurns,  
    secondaryVoltage;
```

```
    printf("Enter primary voltage: ");
```

```
    scanf("%f", &primaryVoltage);
```

```
    printf("Enter primary turns: ");
```

```
    scanf("%f", &primaryTurns);
```

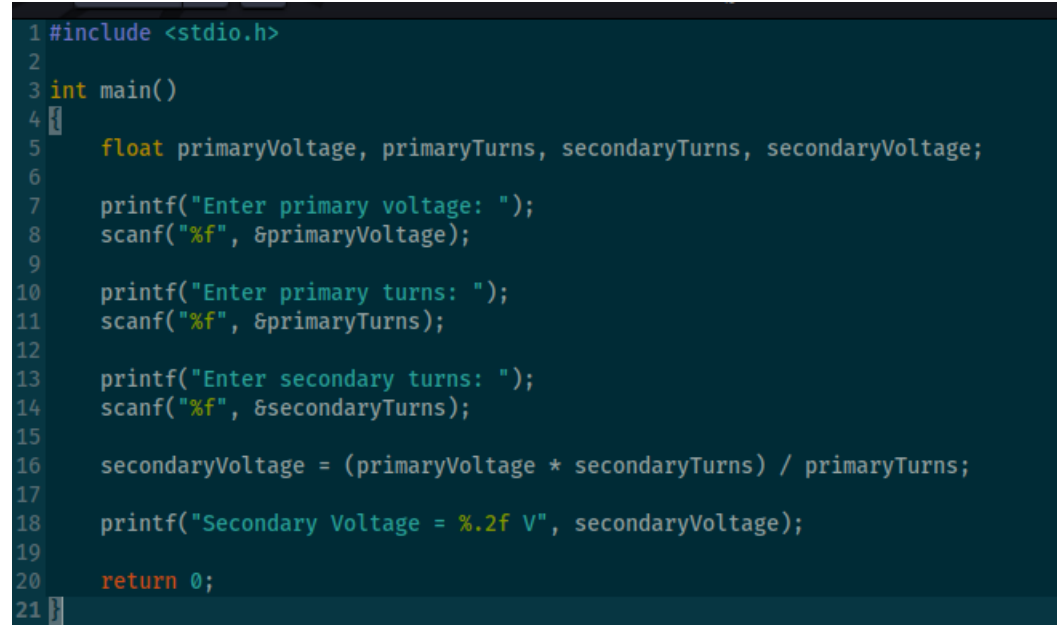
```
    printf("Enter secondary turns: ");
```

```
    scanf("%f", &secondaryTurns);
```

```
    secondaryVoltage = (primaryVoltage * secondaryTurns) /  
    primaryTurns;
```

```
    printf("Secondary Voltage = %.2f V", secondaryVoltage);  
  
    return 0;  
  
}
```

Now we will compile the code and execute it:

A screenshot of a code editor with a dark blue background and light-colored text. The code is a C program that calculates the secondary voltage of a transformer based on primary voltage and turns. The code is numbered from 1 to 21 on the left side. The code includes the standard input/output header, defines the main function, declares variables for primary and secondary voltage and turns, prompts the user for input, reads the input, calculates the secondary voltage, and prints the result.

```
1 #include <stdio.h>  
2  
3 int main()  
4 {  
5     float primaryVoltage, primaryTurns, secondaryTurns, secondaryVoltage;  
6  
7     printf("Enter primary voltage: ");  
8     scanf("%f", &primaryVoltage);  
9  
10    printf("Enter primary turns: ");  
11    scanf("%f", &primaryTurns);  
12  
13    printf("Enter secondary turns: ");  
14    scanf("%f", &secondaryTurns);  
15  
16    secondaryVoltage = (primaryVoltage * secondaryTurns) / primaryTurns;  
17  
18    printf("Secondary Voltage = %.2f V", secondaryVoltage);  
19  
20    return 0;  
21 }
```




Session Actions Edit View Help

```
(kali㉿kali)-[~]  
$ gedit hdoc7.3.c
```

```
** (gedit:38118): WARNING **: 05:00:09.159: Could not
```

```
** (gedit:38118): WARNING **: 05:00:09.160: Could not  
gedit: hdoc7.3.c
```

```
(kali㉿kali)-[~]  
$ cc -g hdoc7.3.c
```

```
(kali㉿kali)-[~]  
$ ./a.out hdoc7.3.c
```

```
Enter primary voltage: 220
```

```
Enter primary turns: 500 ** (gedit:38118): WARNING **: Could not
```

```
Enter secondary turns: 250
```

```
Secondary Voltage = 110.00 V ** (gedit:38118): WARNING **: Could not
```

```
(kali㉿kali)-[~]  
$ ./a.out
```

```
Enter primary voltage: 230
```

```
Enter primary turns: 1000
```

```
Enter secondary turns: 500
```

```
Secondary Voltage = 115.00 V
```

```
(kali㉿kali)-[~]  
$ ./a.out
```

```
Enter primary voltage: 240
```

```
Enter primary turns: 800
```

```
Enter secondary turns: 400
```

```
Secondary Voltage = 120.00 V
```